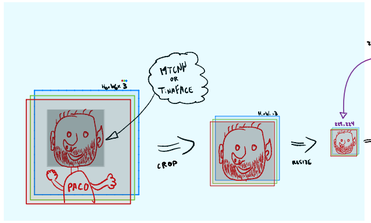
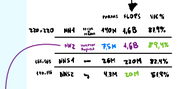




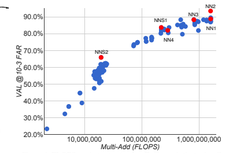
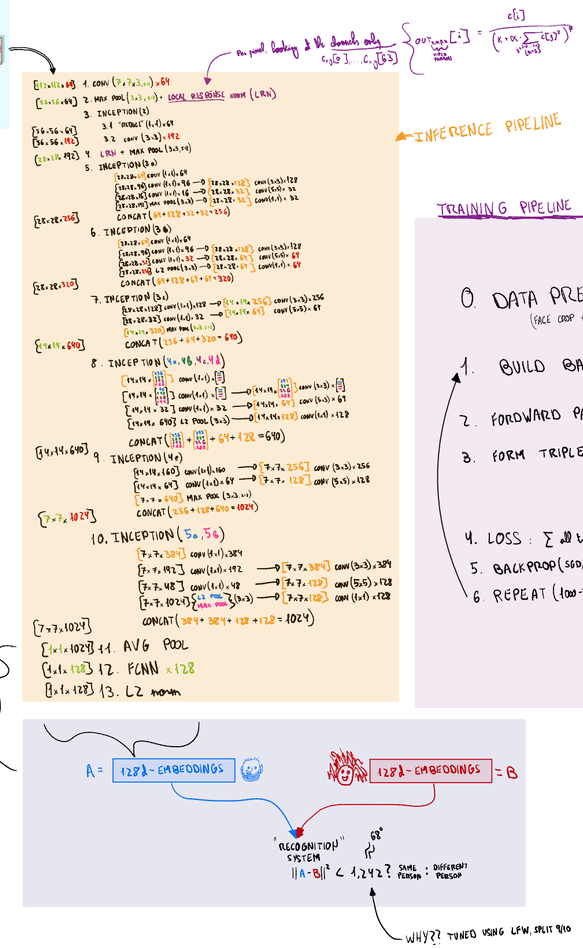
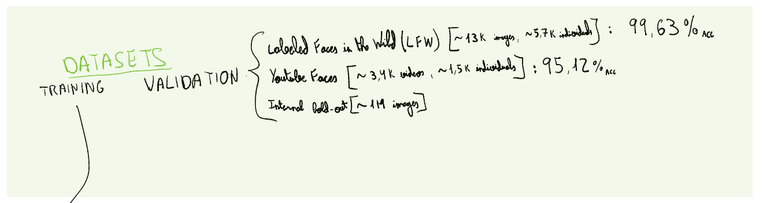


Figure 4. **FLOPS vs. Accuracy trade-off.** Shown is the trade-off between FLOPS and accuracy for a wide range of different model sizes and architectures. Highlighted are the four models that we focus on in our experiments.





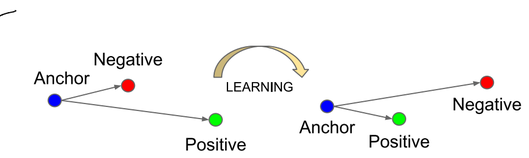
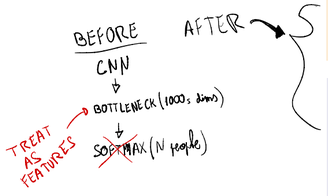


Figure 3. The **Triplet Loss** minimizes the distance between an *anchor* and a *positive*, both of which have the same identity, and maximizes the distance between the *anchor* and a *negative* of a different identity.



ABLATIONS

#training images	VAL
2,600,000	76.3%
26,000,000	85.1%
52,000,000	85.1%
260,000,000	86.2%

Table 6. **Training Data Size.** This table compares the performance after 700h of training for a smaller model with 96x96 pixel inputs. The model architecture is similar to NN2, but without the 5x5 convolutions in the Inception modules.

jpeg q	val-rate
10	67.3%
20	81.4%
30	83.9%
50	85.5%
70	86.1%
90	86.5%

#pixels	val-rate
1,600	37.8%
6,400	79.5%
14,400	84.5%
25,600	85.7%
65,536	86.4%

Table 4. **Image Quality.** The table on the left shows the effect on the validation rate at 10E-3 precision with varying JPEG quality. The one on the right shows how the image size in pixels effects the validation rate at 10E-3 precision. This experiment was done with NN1 on the first split of our test hold-out dataset.

#dims	VAL
64	86.8% ± 1.7
128	87.9% ± 1.9
256	87.7% ± 1.9
512	85.6% ± 2.0

Table 5. **Embedding Dimensionality.** This Table compares the effect of the embedding dimensionality of our model NN1 on our held-out set from section 4.1. In addition to the VAL at 10E-3 we also show the standard error of the mean computed across five splits.

This is a very serious presentation.